

Binary Operations and Algebraic Structures

Ali Karbala

Lebanese University

Reference: Theory of Statistics, James E. Gentle

Outline

- 1 Groups
- 2 Fields
- 3 Vector (or Linear) Space

Binary Operations (Definition)

- A **binary operation** on a set S is a rule for combining two elements of S to form another element.
- If $x, y \in S$, and the operation is denoted by $*$, then the result is written:

$$x * y$$

Properties of a Binary Operation

closure We say S is closed wrt $*$ iff

$$x, y \in S \Rightarrow x * y \in S$$

commutativity We say $*$ defined in S is commutative iff

$$x, y \in S \Rightarrow x * y = y * x$$

associativity We say $*$ defined in S is associative iff

$$x, y, z \in S \Rightarrow x * (y * z) = (x * y) * z$$

Example: Matrix Multiplication

Definition

Let $M_n(\mathbb{R})$ be the set of all $n \times n$ real matrices. Define $*$: $M_n(\mathbb{R}) \times M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$ by

$$A * B = AB \quad (\text{matrix multiplication}).$$

Properties

- **Closure:** $A, B \in M_n(\mathbb{R}) \implies AB \in M_n(\mathbb{R})$.
- **Associativity:** $(AB)C = A(BC)$.
- **Non-commutativity:** In general, $AB \neq BA$.

Groups

One of the most useful algebraic structures is a **group**, which is a set and a binary operation $(S, *)$ with special properties.

Definition 0.0.2 (Group). Let S be a nonempty set and let $*$ be a binary operation. The structure $(S, *)$ is called a group if the following conditions hold:

- ① **Closure:** $x_1, x_2 \in S \Rightarrow x_1 * x_2 \in S$.
- ② **Identity:** $\exists e \in S$ such that $\forall x \in S, e * x = x$.
- ③ **Inverse:** $\forall x \in S, \exists x^{-1} \in S$ such that $x^{-1} * x = e$.
- ④ **Associativity:**

$$x_1, x_2, x_3 \in S \Rightarrow x_1 * (x_2 * x_3) = (x_1 * x_2) * x_3.$$

If a group contains only one element, it is called a **trivial group**, which is not of much interest.

Notes on Groups

- The binary operation need not be commutative, but some specific pairs commute under the operation. In particular, we can easily see that

$$x \circ e = e \circ x \quad \text{and} \quad x \circ x^{-1} = x^{-1} \circ x.$$

- We can also see that e is unique.
- For a given x , the inverse x^{-1} is unique.

Abelian Group

If the binary operation in the group $(S, \circ)$ is commutative, then the group is called a commutative group or an Abelian group.

Example 0.0.4 Group of bijections

Let X be a set and let G be a set of bijective functions g on X . For $g_1, g_2 \in G$, let $g_1 \circ g_2$ represent function composition; that is, for $x \in X$,

$$(g_1 \circ g_2)(x) = g_1(g_2(x)).$$

Group of bijections (continued)

We see that

- $g_1 \circ g_2$ is a bijection on X , so G is closed with respect to $\circ$;
- the function $g_e(x) = x$ is a bijection, and $\forall g \in G, g_e \circ g = g$ (identity);
- $\forall g \in G, \exists g^{-1} \in G$ such that $g^{-1} \circ g = g_e$ (inverse);
- $\forall g_1, g_2, g_3 \in G, g_1 \circ (g_2 \circ g_3) = (g_1 \circ g_2) \circ g_3$ (associativity); that is, $\forall x \in X$, either grouping of the operations is $g_1(g_2(g_3(x)))$.

Therefore, $(G, \circ)$ is a group. The group of bijections in general is not Abelian.

Subgroups

Any subset of the set on which the group is defined that is closed and contains the identity and all inverses forms a group with the same operation as the original group. This subset together with the operation is called a **subgroup**.

We use the standard terminology of set operations for operations on groups.

Generated Groups

A set G_1 together with an operation $\circ$ defined on G_1 generates a group G that is the smallest group $(G, \circ)$ such that $G_1 \subseteq G$.

If G_1 and G_2 are groups over G_1 and G_2 with a common operation $\circ$, the group generated by G_1 and G_2 is $(G, \circ)$, where G is the smallest set containing G_1 and G_2 so that $(G, \circ)$ is a group. Notice that the G may contain elements that are in neither G_1 nor G_2 .

Homomorphisms

It is often of interest to consider relationships between two groups. The simplest and most useful is a morphism, or homomorphism.

Given two groups $G = (G, \circ)$ and $G' = (G', *)$, a homomorphism from G to G' is a function $f : G \rightarrow G'$ such that for $g_1, g_2 \in G$,

$$f(g_1 \circ g_2) = f(g_1) * f(g_2).$$

Often in applications, G and G' are sets of the same kind of objects, say functions for example, and the operations $\circ$ and $*$ are the same, such as function composition.

Isomorphisms

If the homomorphism from G to G' is a bijection, then the homomorphism is called an **isomorphism**.

Since it has an inverse, we say the two groups are **isomorphic**.

Isomorphic groups of transformations are the basic objects underlying the concept of **equivariant** and **invariant** statistical procedures.

Distributivity and Fields

Often it is useful to define two different binary operators, say $+$ and $\cdot$ over the same set. An important type of relationship between the two operators is called distributivity:

$$x, y, z \in S \Rightarrow x \cdot (y + z) = (x \cdot y) + (x \cdot z) \in S.$$

Another very useful algebraic structure is a **field**, which is a set and two binary operations $(S, +, \cdot)$ with special properties.

Definition 0.0.3 (Field)

Let S be a set with two distinct elements and let $+$ and $\cdot$ be binary operations. The structure $(S, +, \cdot)$ is called a field if the following conditions hold:

(f1) $x_1, x_2 \in S \Rightarrow x_1 + x_2 \in S$ (closure of $+$).

(f2) $\exists 0 \in S \ \forall x \in S, 0 + x = x$ (identity for $+$).

(f3) $\forall x \in S \ \exists -x \in S$ s.t. $x + (-x) = 0$ (inverse wrt $+$).

(f4) $x_1 + x_2 = x_2 + x_1$ (commutativity of $+$).

(f5) $x_1 + (x_2 + x_3) = (x_1 + x_2) + x_3$ (associativity of $+$).

Definition 0.0.3 (Field)

(f6) $x_1, x_2 \in S \Rightarrow x_1 \cdot x_2 \in S$ (closure of $\cdot$).

(f7) $\exists 1 \in S \ \forall x \in S, 1 \cdot x = x$ (identity for $\cdot$).

(f8) $\forall x \neq 0 \in S \ \exists x^{-1} \in S$ s.t. $x^{-1} \cdot x = 1$ (inverse wrt $\cdot$).

(f9) $x_1 \cdot x_2 = x_2 \cdot x_1$ (commutativity of $\cdot$).

(f10) $x_1 \cdot (x_2 \cdot x_3) = (x_1 \cdot x_2) \cdot x_3$ (associativity of $\cdot$).

(f11) $x_1 \cdot (x_2 + x_3) = (x_1 \cdot x_2) + (x_1 \cdot x_3)$ (distributivity).

Remarks on Fields

Although a field is a structure consisting of a set and two binary operations, $(S, +, \cdot)$, we often refer to the field simply by the name of the set, S .

Notice that the word “field” is also used in a slightly different sense to refer to a structure consisting of a collection of sets and the unary operation of set complementation and the binary operation of set union (we will see this later in Measure Theory).

Rings

- A ring is a structure having all properties of a field except, possibly, no inverses with respect to the second operation (lacking property (f8)), and possibly having only one element (a ring can be trivial).
- Sometimes a ring in which the second operation is commutative (as we have required and as is always required for a field) is called a **commutative ring**.
- A structure that may possibly lack that commutativity is called simply a **ring**.
- The concept of homomorphisms between fields or between rings follows similar development as that of homomorphisms between groups.

Examples: Rings and Fields

- $(\mathbb{Z}, +, \cdot)$ is a **ring**, but not a field (no multiplicative inverses for most elements, e.g. $2^{-1} \notin \mathbb{Z}$).
- $(\mathbb{Q}, +, \cdot)$ is a **field** (every nonzero rational has an inverse, e.g. $\frac{2}{3}^{-1} = \frac{3}{2}$).
- Other common fields:
 - $\mathbb{R}$ (reals: all nonzero numbers invertible).
 - $\mathbb{C}$ (complex numbers: all nonzero numbers invertible).

Finite Fields

- Fields with a finite number of elements are called **Galois fields**.
- Denoted $\text{GF}(m)$ or $\mathbb{F}(m)$, where $m =$ number of elements.
- Example: $\text{GF}(5)$
 - Set $\{0, 1, 2, 3, 4\}$.
 - Addition $+$ with identity 0.
 - Multiplication $\cdot$ with identity 1.
- Operations defined by **addition and multiplication**.

Ordered Fields

We have defined ordered sets in Section 0.0.1 in terms of the existence of a binary relation. We now define ordered fields in terms of the field operations.

Definition 0.0.4 (ordered field) A field S is said to be **ordered** if there is a subset P of S such that

- $P \setminus (-P) = \emptyset$;
- $P \cup \{0\} \cup (-P) = S$;
- $x, y \in P \Rightarrow x + y, x \cdot y \in P$.

Archimedean Ordered Fields

Definition 0.0.5 (Archimedean ordered field) An ordered field S defined by the subset P is said to be **Archimedean ordered** if for all $x \in S$ and all $y \in P$, there exists a positive integer n such that $ny > x$.

An Archimedean ordered field must be dense, in the sense that we can find an element between any two given elements.

Theorem 0.0.3 Let S be an Archimedean ordered field, and let $x, y \in S$ such that $x < y$. Then there exists integers m and n such that $m/n \in S$ and

$$x < \frac{m}{n} < y.$$

Definition 0.0.6 (linear space)

A structure $(V, +)$ in which the operator $+$ is commutative and associative, and for which the multiplication of a real number a and an element $x \in V$ is defined as a closed operation in V (denoted ax), is called a **linear space** if for any $x, y \in V$ and any $a, b \in \mathbb{R}$:

- $a(x + y) = ax + ay$,
- $(a + b)x = ax + bx$,
- $(ab)x = a(bx)$,
- $1x = x$, where 1 is the multiplicative identity in $\mathbb{R}$.

Linear Combinations, Linear Independence, and Basis Sets

- Given $x_1, x_2, \dots \in V$ and $c_1, c_2, \dots \in \mathbb{R}$, $\sum_i c_i x_i$ is called a **linear combination**.
- A set of elements $x_1, x_2, \dots \in V$ are said to be **linearly independent** if $\sum_i c_i x_i = 0$ for $c_1, c_2, \dots \in \mathbb{R}$ implies that $c_1 = c_2 = \dots = 0$.
- Given a linear space V and a set $B = \{b_i\}$ of linearly independent elements of V , if for any element $x \in V$, there exist $c_1, c_2, \dots \in \mathbb{R}$ such that $x = \sum_i c_i b_i$, then B is called a **basis set** of V .
- A set of elements $x_1, \dots, x_d \in V$ are said to be **affinely independent** if $x_2 - x_1, \dots, x_d - x_1$ are linearly independent. Affine independence in this case insures that the points do not lie in a set of less than $d - 1$ dimensions.

Inner Products

Definition 0.0.7 (inner product)

If V is a linear space, an inner product on V is a real-valued function, denoted by $\langle x, y \rangle$ for all x and y in V , that satisfies the following three conditions for all x, y , and z in V :

- 1 Nonnegativity and mapping of the identity: if $x \neq 0$, then $\langle x, x \rangle > 0$ and $\langle 0, x \rangle = \langle x, 0 \rangle = \langle 0, 0 \rangle = 0$.
- 2 Commutativity: $\langle x, y \rangle = \langle y, x \rangle$.
- 3 Factoring of scalar multiplication in inner products: $\langle ax, y \rangle = a\langle x, y \rangle$ for real a .

Inner Products (cont.)

- ④ Relation of vector addition to addition of inner products:

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle.$$

Inner products are often called dot products, although “dot product” is often used to mean a specific inner product.

A linear space together with an inner product, $(V, \langle \cdot, \cdot \rangle)$, is called an **inner product space**.

Inner product spaces give rise to the very useful concept of orthogonality. If $(V, \langle \cdot, \cdot \rangle)$ is an inner product space, the elements $x, y \in V$ are orthogonal if $\langle x, y \rangle = 0$. We write $x \perp y$.

Norms

Definition 0.0.8 (norm)

If V is a linear space, a norm on V is a function, denoted by $\| \cdot \|$, from V to $\overline{\mathbb{R}}^+$ that satisfies the following three conditions for all x and y in V :

- 1 Nonnegativity and mapping of the identity: if $x \neq 0$, then $\|x\| > 0$, and $\|0\| = 0$.
- 2 Relation of scalar multiplication to real multiplication:
 $\|ax\| = |a| \|x\|$ for real a .

Norms (cont.)

- ③ Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.

A linear space together with a norm, $(V, \|\cdot\|)$, is called a **normed linear space**.

Normed linear spaces give rise to the very useful concept of **projections**.

Projections

Definition 0.0.9 (projection)

Let $(V, \|\cdot\|)$ be a normed linear space and let $(W, \|\cdot\|)$ be a normed linear subspace; that is, $(W, \|\cdot\|)$ is a normed linear space and $W \subseteq V$.

For $x \in V$, the **projection** of x onto W is x_p such that

$$\|x - x_p\| = \inf_{y \in W} \|x - y\|.$$

Pseudonorms

Pseudonorms

If, in the first condition in Definition 0.0.8, the requirement

$$\text{if } x \neq 0, \text{ then } \|x\| > 0$$

is replaced by

① if $x \neq 0$, then $\|x\| \geq 0$ and $\|0\| = 0$,

the resulting function is called a **pseudonorm**, or a **seminorm**.

Norms Induced by an Inner Product

Theorem 0.0.4

Let $\langle \cdot, \cdot \rangle$ be an inner product on V . Then

$$\sqrt{\langle x, x \rangle}, \quad x \in V,$$

is a norm on V .

Metrics Induced by a Norm

Theorem 0.0.5

Let $\|\cdot\|$ be a norm on V . Then

$$\|x - y\|, \quad x, y \in V,$$

is a metric on V .

The terms “**normed linear space**” and “**linear metric space**” are therefore equivalent, and we will use them interchangeably.

Cauchy Sequences and Completeness

Let $A = \{x_i\}$ be a countable sequence of elements of a linear metric space with metric $d(\cdot, \cdot)$.

If for every $\varepsilon > 0$, there exists a constant n_ε such that

$$d(x_n, x_m) < \varepsilon \quad \forall m, n > n_\varepsilon, \quad (0.0.27)$$

then A is called a **Cauchy sequence**.

Notice that the definition of a Cauchy sequence depends on the metric.

A sequence of elements of a linear metric space converges only if it is a Cauchy sequence.

A normed linear space is said to be **complete** if every Cauchy sequence in the space converges to a point in the space.

Banach and Hilbert Spaces

A complete normed linear space is called a **Banach space**.

A Banach space whose metric arises from an inner product is called a **Hilbert space**.

Equivalently, a Hilbert space is the same as a complete inner product space.

Completion of a Metric Space

Given a metric space (X, d) , the space (X^c, d^c) is called the **completion** of (X, d) if

- X is dense in X^c
- $d^c(x, y) = d(x, y) \quad \forall x, y \in X$

For any given metric space, the completion exists.